Experiment 9: Travelling Salesman Problem (TSP)

Aim

To write a Python program to solve the Travelling Salesman Problem (TSP) using brute-force method.

Procedure

1. Import the permutations function.
2. Define the distance matrix.
3. Generate all possible paths.
4. Calculate the total cost of each path.
5. Find the minimum cost path.
6. Display the shortest route and its cost.

Source Code

main.py

Visualize

Run ▶

```python
from itertools import permutations

graph = [
    [0, 10, 15, 20],
    [10, 0, 35, 25],
    [15, 35, 0, 30],
    [20, 25, 30, 0]
]

cities = [0, 1, 2, 3]

min_cost = float('inf')
best_path = None

for path in permutations(cities[1:]):
    current_path = (0,) + path
    cost = 0

    for i in range(len(current_path)-1):
        cost += graph[current_path[i]][current_path[i+1]]

    cost += graph[current_path[-1]][0]

    if cost < min_cost:
        min_cost = cost
        best_path = current_path

print("Best Path:", best_path + (0,))
print("Minimum Cost:", min_cost)
```

Output

```
Best Path: (0, 1, 3, 2, 0)
Minimum Cost: 80
```

Result

Thus, the Travelling Salesman Problem was solved successfully using the brute-force approach.